

Artificial Intelligence for Cybersecurity in Cloud Computing

(English Executive Summary)

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Executive Summary

This book presents a structured academic study on the role of Artificial Intelligence (AI) in improving cybersecurity within cloud computing environments. It aims to connect foundational security principles with modern AI-driven solutions in real-world cloud infrastructures.

Cloud computing has transformed the way organizations manage data, applications, and digital services. Service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) provide flexibility and scalability. However, they also introduce new security challenges, including shared responsibility issues, identity management risks, API vulnerabilities, data breaches, distributed attacks, and insider threats.

The book begins by explaining the fundamental concepts of cloud architecture and cybersecurity, including the CIA triad (Confidentiality, Integrity, and Availability), threat modeling, and risk management frameworks. It then introduces core machine learning and deep learning concepts necessary for understanding AI-based security systems.

A major part of the book focuses on applying AI techniques to cloud security problems. It reviews supervised, unsupervised, and reinforcement learning approaches and discusses how models such as Decision Trees, Random Forest, Support Vector Machines (SVM), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), LSTM, GAN, and Transformer-based architectures can be used for:

- Intrusion Detection Systems (IDS)
- Anomaly detection in cloud traffic
- Malware classification
- User and entity behavior analytics (UEBA)
- Security Information and Event Management (SIEM) enhancement
- Automated response systems (SOAR)

The book also examines practical implementation aspects in major cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). It highlights how AI models can support predictive security, real-time monitoring, and automated threat response in multi-cloud and hybrid environments.

In addition to technical solutions, the book addresses important operational challenges. These include data quality limitations, scalability constraints, model explainability (Explainable AI), adversarial attacks, privacy regulations, concept drift in dynamic systems, and the gap between laboratory evaluation and real-world deployment.

The final chapters discuss emerging trends such as federated learning for privacy-preserving security, AI in serverless architectures, multi-cloud security strategies, and the impact of quantum computing on encryption and threat detection.

Overall, this book provides a balanced academic and practical framework for researchers, cloud engineers, cybersecurity professionals, and IT managers who seek to apply AI methods responsibly and effectively in cloud environments. The focus is on clarity, structured analysis, and realistic deployment considerations rather than theoretical abstraction.

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